

AI-Powered Credit Risk Intelligence Platform

End-to-End Explainable Underwriting & Conversational
Data Stack

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The Business Use Case

THE PROBLEM

Banks face immense pressure to process loan applications rapidly and accurately. Legacy systems fail to parse extensive relational transactional histories and lack the transparency needed to pass regulatory compliance audits.

THE SOLUTION: INTEGRATED PLATFORM



Risk Identification

Identify high-risk loan applicants early using multi-table features.



Automated Decisions

Decision support via ML bounded by strict policy rules.



Analytical Queries

Plain-English query windows for non-technical risk analysts.

Data Processing & Architectural Ingestion

DATA FOOTPRINT: 13M+ SOURCE ROWS

applications

307,511

Core demographics and financials.

bureau

1.71M

External history with other institutions.

prev_app

1.67M

Internal historical loan applications.

installments

13.60M

Micro-level payment/delinquency.

STORAGE TECHNOLOGY



Parquet + DuckDB

Optimized for high-performance local analytical processing.

ENGINEERING RATIONALE

Converting text CSVs into compressed binary **Parquet** files drastically lowered memory overhead.

This enables **DuckDB** to run ultra-fast analytical queries directly on local environments.

Exploratory Data Analysis (EDA) Metrics

ANALYSIS PERFORMED ON THE DATASET

Target Profiling

Calculated the base default rate to isolate class distribution.

Categorical Risk Analysis

Cross-tabulated employment types against loan default percentages.

Financial Distribution Profiling

Evaluated the debt-burden spread by dividing total loan credit by annual applicant income.

Data Quality Checks

Tracked missing values and exposed anomalies (e.g., retired group with error code **365,243** days employed).

Executive EDA Dashboard

Tab 1 — Executive Business Insights Screen



Class Imbalance

Found an explicit **8.07% baseline default rate**, showcasing a heavy 11:1 data skew.



Income Type Skew

Unemployed applicants (**38.1%**) and those on maternity leave (**35.0%**) show massive default concentrations.

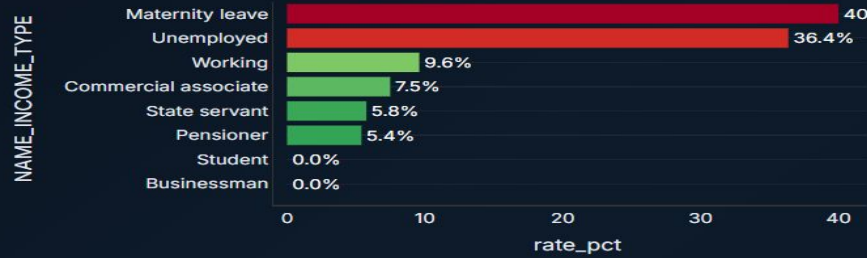


Financial Averages

Baseline loan credit averages **\$599,026** against an average income of **\$168,798**.

Tab 1 — Executive EDA & Business Insights UI

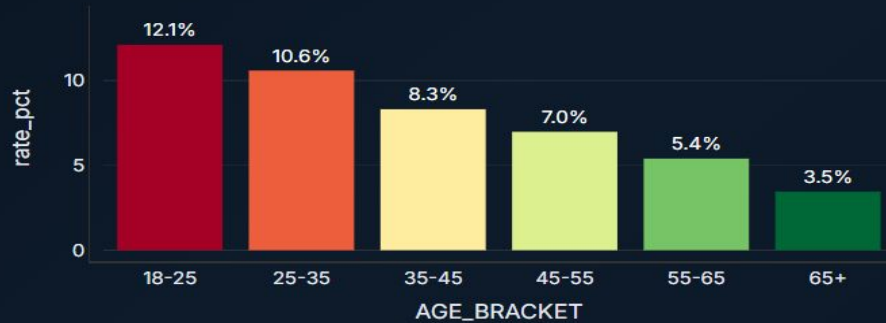
Default Rate by Income Type (%)



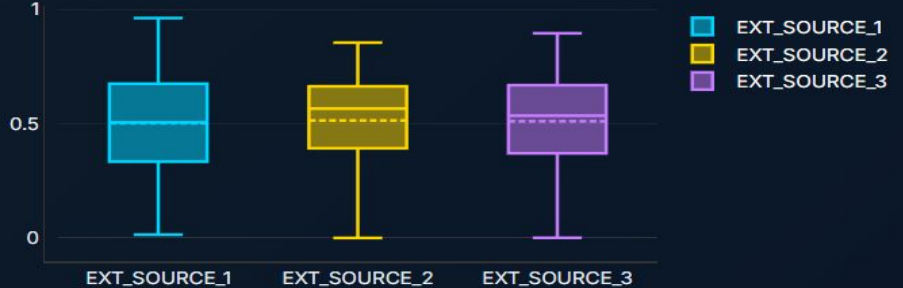
Credit/Income Ratio Distribution



Default Rate by Age Bracket



External Credit Bureau Score Distribution



Tab 1 — Executive EDA & Business Insights UI

Data Quality — Top 20 Columns by Missing Value Rate



Talk-to-Data Architecture

Conversational Chatbot & Text-to-SQL Pipeline

Technology Layer Used

Meta Llama-3.1-8b-instant Model (via Groq Inference Gateway)

Why This Framework

Groq provides an OpenAI-compatible API bridge with lightning-fast speeds on a free tier, satisfying cost constraints while delivering live GenAI capabilities.

Token Optimization Design

Passes a lightweight metadata schema dictionary in the system prompt instead of full tables. This minimizes token consumption and prevents crashes.

Conversational Chatbot Live Interface

Tab 4 — Live Talk-to-Data Chatbot Screen



Functional Flow

User Natural Language Question → LLM SQL Translation → Engine Execution (credit_risk.duckdb) → Human Response



Live Query Verification

Input: *"Show me the average income of applicants who defaulted"*

Result: Instant background pipeline trigger returns a baseline mean of **\$165,612**.

Tab 4 — Talk-to-Data Conversational UI

Top 10 applicants by credit amount

☹️ **Returns the 10 applicants with the highest credit amounts, along with their income and default status.**

Based on our analysis, the top 10 applicants with the highest credit amounts are all requesting approximately \$4.05 million in credit. Notably, 9 out of these 10 applicants have a credit history, as indicated by a TARGET value of 0, suggesting they are not defaulting on their payments. The only exception is applicant 315893, who has a TARGET value of 1, indicating a potential default risk.

	SK_ID_CURR	AMT_CREDIT	AMT_INCOME_TOTAL	credit_income_ratio	TARGET
0	433294	4050000	405000	10	0
1	434170	4050000	450000	9	0
2	238782	4050000	931365	4.35	0
3	240007	4050000	587250	6.9	0
4	255139	4050000	1307587.5	3.1	0

▼ SQL Query

```
SELECT SK_ID_CURR, AMT_CREDIT, AMT_INCOME_TOTAL, round(AMT_CREDIT / NULLIF(AMT_INCOME_TOTAL, 0), 2) AS credit_income_rati
```

Machine Learning Risk Layer

Machine Learning Pipeline & Training Rationale

Algorithm Used

LightGBM Classifier (Light Gradient Boosting Machine)

Why This Algorithm

Tree-based boosting outperforms deep learning on tabular files. It optimizes speed, scales to millions of rows, and handles missing blocks natively using histogram-based splits.

Class Imbalance Mitigation

Uses Stratified 5-Fold CV. Tuning factor (`scale_pos_weight = 11.39`) penalizes default errors more heavily to correct the 8.07% data skew.

Operational Outputs: Actionable **Risk Score (0-100)** & Structural **Risk Band (Low / Med / High)**

Explainable AI Layer

Tab 2 — Underwriting Decision Support & SHAP Core

Technology Used

SHAP (SHapley Additive exPlanations) Waterfall Engine

Why This Framework

Breaks down black-box models to satisfy legal auditing mandates. It computes local feature attributions to show precisely which criteria pushed a specific applicant's default score up or down.

Interactive Simulation Mode

Provides manual sliders for loan officers to instantly test and run alternative risk profiles (Income, Credit Amount, external bureau flags).

Tab 2 — Underwriting Decision & Explainability UI

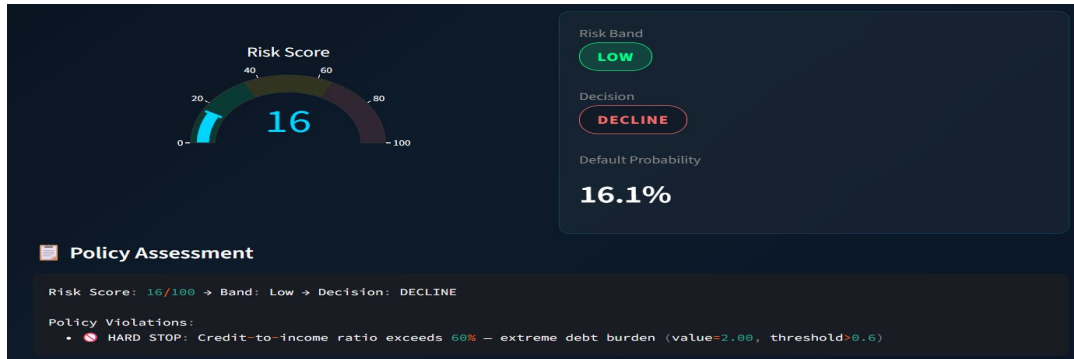
Enter Applicant Features

Input mode

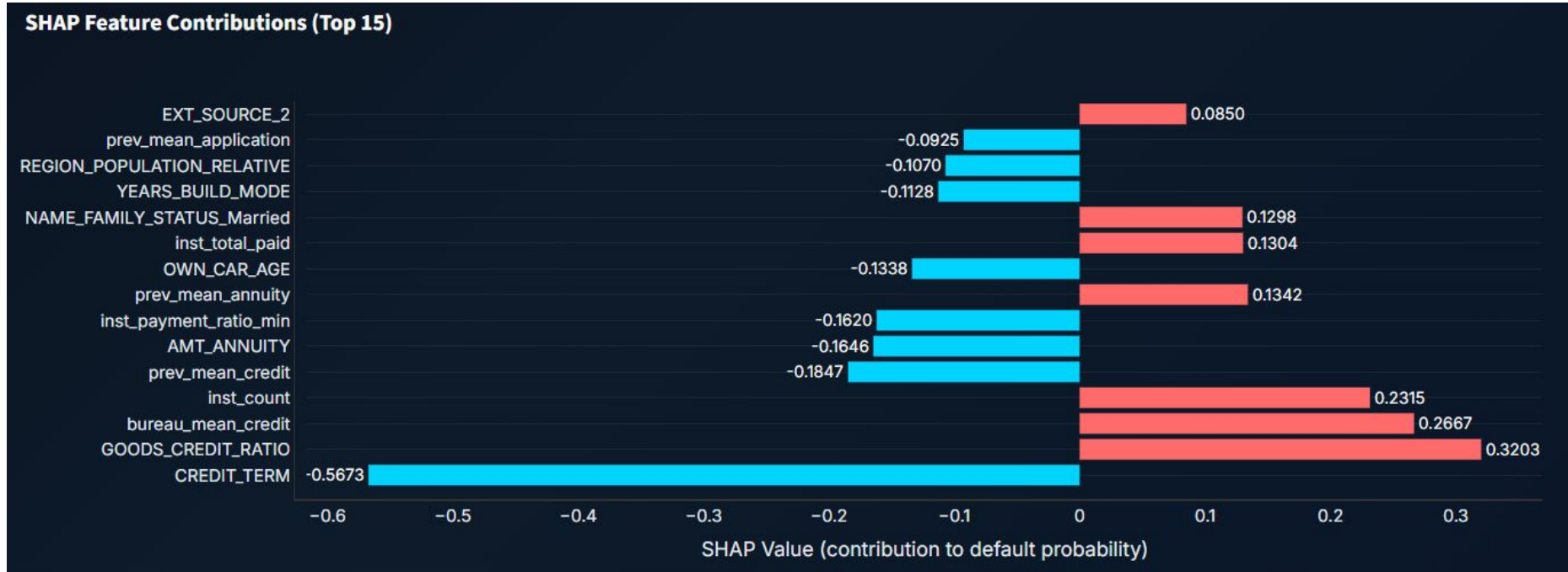
☒ Manual Entry ☐ Use Dataset Row

Annual Income (\$)	EXT_SOURCE_1	Age (years)
150000 - +		
Credit Amount (\$)	EXT_SOURCE_2	Years Employed
300000 - +		
Annuity (\$)	EXT_SOURCE_3	Bureau Max Overdue Days
15000 - +		

 Predict Risk



Tab 2 — Underwriting Decision & Explainability UI



Credit Policy Rule Engine

Tab 3 — Operational Risk Guardrails & Rule Derivation

The Compliance Need

In commercial banking, probabilistic ML models are not allowed to operate unchecked. They must be bounded by deterministic policy guardrails to enforce hard compliance limits.

Layer 1 Hard-Stops Implemented: Automatic Decline Flags

CREDIT_INCOME_RATIO > 0.6: Restricts high debt burdens relative to income.

EXT_SOURCE_MEAN < 0.15: Screens out severe historical credit bureau failures.

bureau_max_overdue_days > 180: Automatically declines active, long-term delinquencies.

Containerized Deployment

Production Deployment via Docker & Compose

Technology: Docker + Docker Compose Engine

Why This Framework

Satisfies requirements for a single-command build step, eliminating environment discrepancies for the grading evaluator.

Orchestration Metrics

Dockerfile: Uses `python:3.11-slim` base, configures `PYTHONPATH`, and maps port `8501`.

docker-compose.yml: Maps shared database volumes to ensure heavy datasets persist safely during build cycles.

Deployment Command: `docker-compose up --build`

System Performance Metrics Matrix

Model Training Evaluation: Stable Performance & Risk Stratification

Key Performance Metrics Summary

ROC-AUC

Measures the model's ability to distinguish between classes. Values indicate strong predictive power.

Gini Coefficient

Derived from the AUC, it quantifies the inequality of the distribution for risk grading.

KS-Statistic

Measures the maximum separation between default and non-default cumulative distributions.

Brier Score

Assesses the accuracy of probabilistic predictions (lower scores indicate better calibration).

Evaluation Matrix

<i>Performance Metric</i>	<i>Achieved Project Value</i>	<i>Target Baseline</i>	<i>Analysis & Significance</i>
ROC-AUC	0.7797	> 0.70	Highly accurate separation of defaulting loan profiles.
Gini Coefficient	0.5593	> 0.40	Strong rank-order stability; confirms commercial viability.
KS-Statistic	0.4233	> 0.40	Excellent capability to classify risk into clear deciles.
Brier Score	0.1694	Closer to 0.0	High probability calibration; predictions match real default frequencies.

Thank you!